Thermia Duo heat pump



Thermia Duo

Installation instructions

9680–54961001
Version 4

Contents

1 Important information 4
2 Placement, unpacking and installation 4
2.1 Preparing the installation site 5
2.2 Unpacking and setting up 6
2.3 Removing the front panel 6
2.4 Required service space around heat pump . 7
2.4.1 Dimensions 7
3 Dimensions for connections 8
4 Pipe installation 9
4.1 Safety valve 9
4.2 Coolant lines 10
4.3 Heating system supply and return lines . . 10
4.4 Filling the heating system 10
4.5 Wiring diagram for heating system – System VL 11
4.6 Wiring diagram for heating system – System D 12
5 Heat pump components 13
6 Outdoor temperature sensor . . . 14
7 Electrical installation 15
7.1 Electrical connections 15
7.1.1 Connecting the outdoor temperature sensor 19
7.1.2 Connecting the room thermostat (option) . . 19
7.1.3 Connecting the supply line sensor 20
7.2 Additional features 24
8 Filling the coolant system 25
9 Preparing for use 27
9.1 Fitting the front panel 28
9.2 Requirements for switching on heating 29
9.2.1 Heat pump (compressor) 29
9.2.2 Additional heating 29
9.3 Requirements for switching off heating . 29
9.3.1 Heat pump (compressor) 29
9.3.2 Additional heating 29
9.4 Requirements for switching on warm water 30
9.4.1 Heat pump (compressor) 30
9.4.2 Additional heating 30
9.5 Requirements for switching off warm water 30
9.5.1 Heat pump (compressor) 30
9.5.2 Additional heating 30
10 Computer control panel 31
10.1 Features 31
10.2 Display information and settings 32
10.2.1 Normal 32
10.2.2 Alarms 34
10.2.3 Customer information 35
10.2.3.1 Operation 35
10.2.3.2 Heat curve 36
10.2.3.3 Heat curve 2 36
10.2.3.4 Temperature 37
10.2.3.5 Operating time 37
10.2.4 Servicing 38
10.2.4.1 Warm water 38
10.2.4.2 Heat pump 38
10.2.4.3 Additional heater 39
10.2.4.4 Manual test 40
10.2.4.5 Installation 1A 41
10.2.4.6 Menu for Thermia Duo Optimum 42
10.3 Control system – available operating modes 43
10.4 Heating system 44
10.4.1 Adjusting the "ROOM" setting 44
10.4.2 To change the "ROOM" setting 44
10.5 Setting the heat curve 45
10.5.1 Adjusting the "CURVE" setting 45
10.5.2 To change the "CURVE" setting 45
10.6 Maximum return line temperature . . 46
10.7 Adjusting the room factor 46
10.8 Warm water production 46
10.8.1 Reading the warm water temperature 46
11 Troubleshooting 48
12 Factory settings for control computer 51
13 Refrigerant 54
14 Technical specifications . . . 55

Thermia Värme AB reserves the right to change parts and specifications without prior notice.



This symbol indicates safety precautions that must be followed. Failure to follow these precautions may endanger life or cause damage to the equipment and its components.

1 Important information

Installation may only be carried out by an authorized installer in accordance with these installation instructions and the relevant rules and regulations.

The heat pump must not be installed where there is a risk of freezing!

The floor must be capable of supporting the total weight of the heat pump (see Technical specifications).

The heat pump must be located on a stable floor surface.

2 Placement, unpacking and installation

Preparing the installation site.

- Move the heat pump into position and use the adjustable feet to ensure that it stands upright. Remove the front panel.
- Decide which side the coolant hose will be connected to.
- Make sure there is sufficient space for installation (see page 5).
- It is recommended that the condensate drain is connected from the drip bowl drain pipe to a floor drain. The drain pipe exits through a hole in the bottom panel. The hose diameter must be 14/10 mm.
- Carry out the pipe installation.
- Fill the heating system and then bleed it.
- Carry out the electrical installation.
- Fill the coolant circuit.
- Adjust the heat pump's factory settings if necessary.
- Carry out a function test.
- Fill in the installation certificate (see page 57) before the end user starts to use the installation.

2.1 Preparing the installation site

- Make holes in the walls for the access pipes for the coolant pipes (1). Follow the dimensional sketches and connection sketches on pages 5–6.
- The outgoing and incoming coolant pipes should have different access holes in the wall.
- Special access holes must be provided if there is a risk of groundwater seeping in.
- Install the access pipes (1) so that they slope slightly downwards towards the outside. The gradient must be at least 1 cm every 30 cm. Cut the pipes at an inward angle (as shown) so that rainwater cannot get into the pipes.
- Make sure that the access pipes are correctly spaced so that there is room for other installations.
- Run the coolant pipes (2) through the access pipes and into the installation room.
- Make good the holes in the walls with mortar (3).
- Make sure that the coolant pipes (2) are centred in the access pipes (1) so that thermal insulation can be applied evenly all round.
- Seal the access pipes (1) with a suitable sealant (foam plastic) (4).

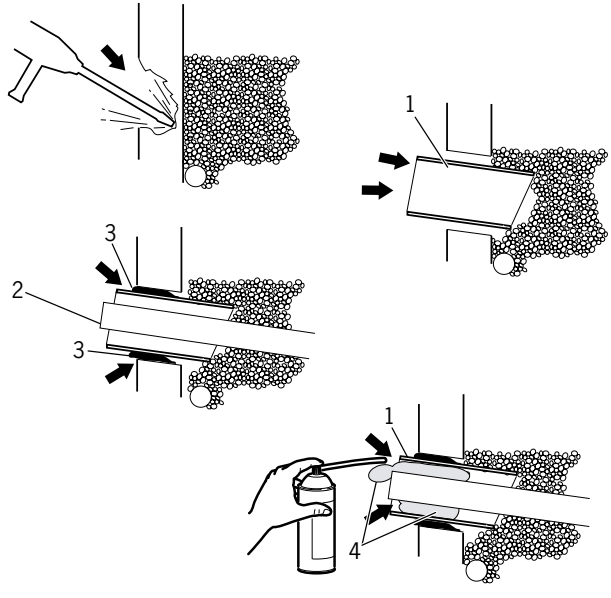


Fig 1. Preparing for installation

2.2 Unpacking and setting up

The heat pump is packaged in plastic film and supplied on a wooden pallet.

- Check that the delivery is complete and that there is no damage.
- Move the heat pump to the installation site.
- Cut the packing straps and remove the plastic film.
- Lift the heat pump off the pallet and adjust it using the adjustable feet (5) so that it stands upright on the floor.

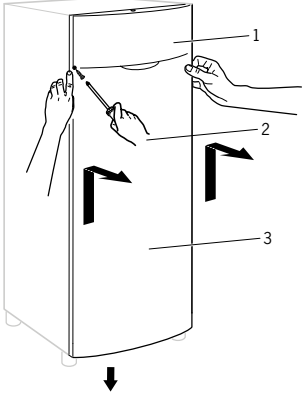


Fig 3. Removing the front panel

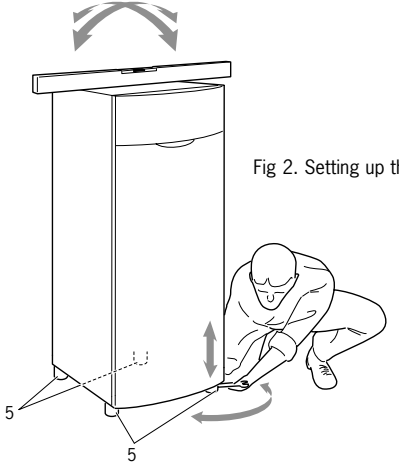


Fig 2. Setting up the heat pump

2.3 Removing the front panel

Remove the front panel as follows:

- Open the hinged top panel (1).
- Remove the screws (2).
- Slide the front panel (3) upwards and lift it off while pulling it towards you.

Take care to avoid damaging the control system wiring when lifting off the front panel.

- Place the front panel next to the heat pump.

2.4 Required service space around heat pump

To simplify installation and subsequent inspection and maintenance there must be sufficient clear space around the heat pump as specified below:

- 300 mm on either side
- 300 mm above
- 600 mm in front
- 10 mm behind

2.4.1 Dimensions

Figure 4 shows the dimensions of the heat pump and the space required for servicing.

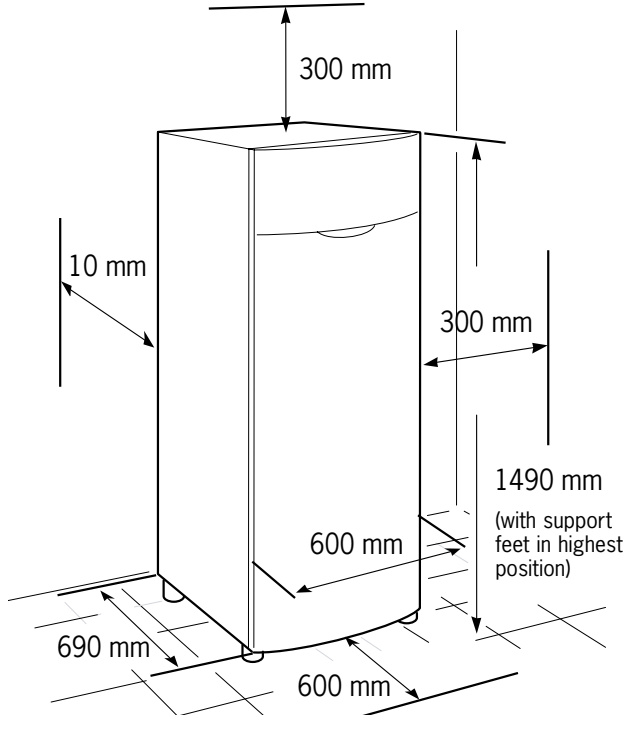


Fig 4. Required service space around heat pump

3 Dimensions for connections

The warm water return line can be connected on the left or right side of the heat pump.

Diameter

Outgoing coolant (from HP)	28 mm copper
Incoming coolant (to HP)	28 mm copper
Heating system supply line	28 mm copper
Heating system return line	28 mm copper
Warm water line from warm water tank	22 mm

Key to symbols:

1. From warm water tank
2. Outgoing coolant (from HP)
3. Incoming coolant (to HP)
4. Heating system supply line
5. Heating system return line
6. Cable exits

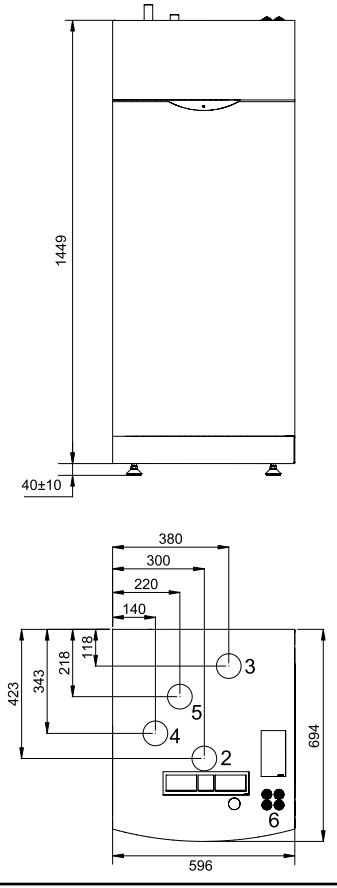


Fig. 5 Dimensions for connections

4 Pipe installation

To avoid leaks, make sure that connecting pipes are not installed under tension!

- Make sure that pipes are installed in accordance with the dimensional sketches and connection sketches on pages 5–6.
- Pipe installation must be carried out by an authorized installer.

Installation must comply with local building regulations, and:

NOTE! It is extremely important that the heating system is bled entirely of air.

NOTE! Bleed valves must be installed where necessary.

4.1 Safety valve

- If the radiator system has a closed expansion tank then this system must also be fitted with an approved pressure gauge and safety valve with a minimum rating of DN 20 and a maximum opening pressure of 1.5 bar.

The heat pump must be connected to an expansion tank and safety valve in accordance with the relevant regulations. The connecting pipe between the tank and safety valve must rise continuously. This means that the pipe must not deviate below an imaginary horizontal line.

4.2 Coolant lines

- Insert the flexible return line (incoming coolant, marked Brine return) through the rubber return line sleeve in the top panel of the heat pump.
- Connect the return line with all associated components.
- Run the flexible supply line (outgoing coolant) through the supply line rubber sleeve in the top panel of the heat pump.
- Connect the supply line with all associated components.
- Cover both lines with thermal insulation that incorporates a vapour barrier.

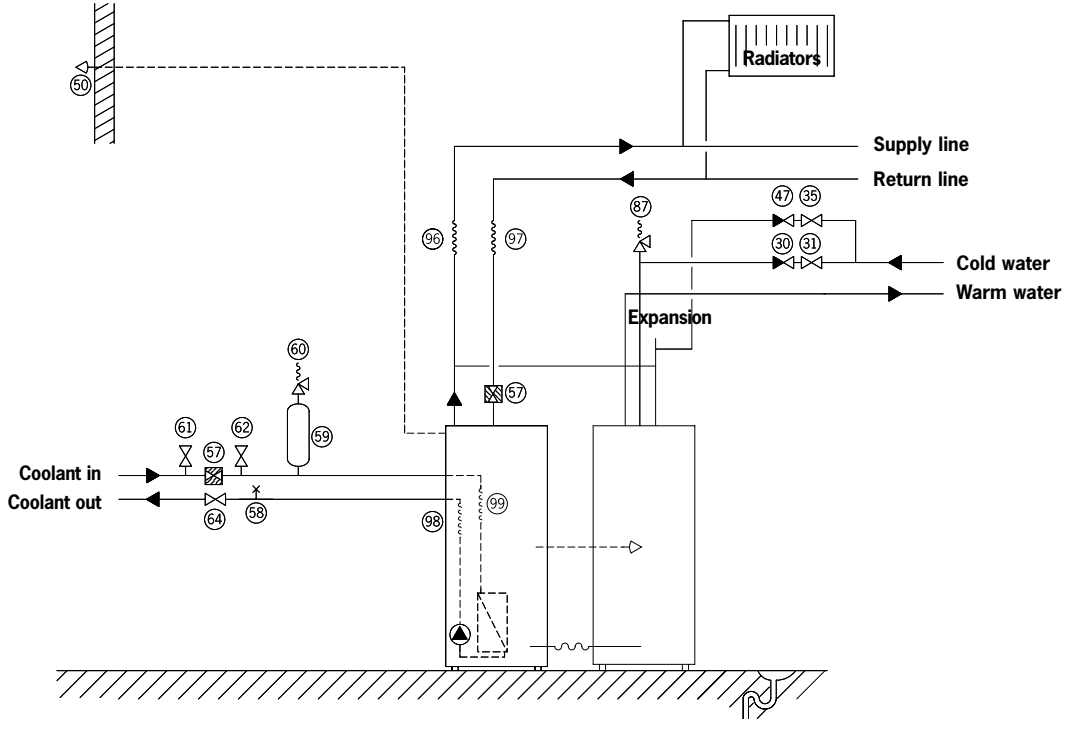
4.3 Heating system supply and return line

- Fit the particle filter as indicated in the system layout

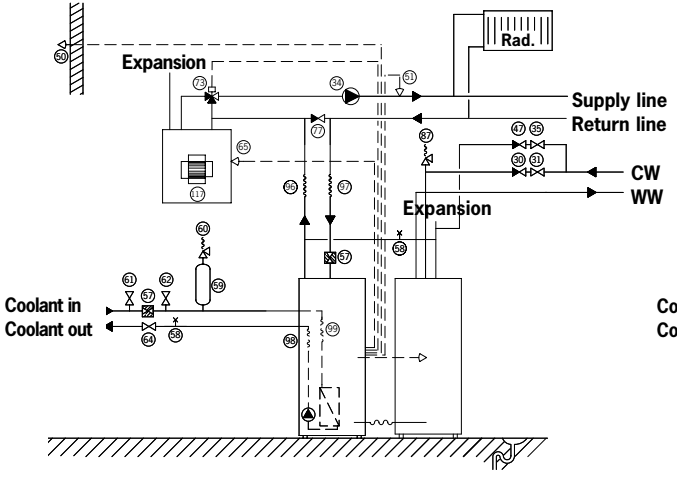
4.4 Filling the heating system

Fill the heating system with water to a pressure of around 1 bar. Then bleed the entire heating system and top up with water.

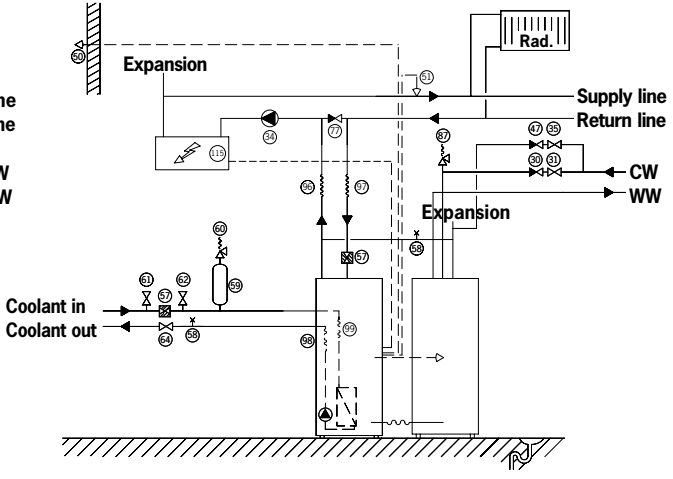
4.5 Wiring diagram for heating system – System VL



4.6 Wiring diagram for heating system – System D



Duo with external shunt option



Duo in parallel with external immersion heater

5 Heat pump components

1. Front panel, removable
2. Instrument panel with controls
3. Switching valve
4. Supply line sensor
5. Circulation pump, heating system
6. Additional heating, electric heater
7. Evaporator, isolated
8. Expansion valve
9. Drying filter
10. Circulation pump, cooling system
11. Return line, heating system, 28 mm Copper
12. Supply line, heating system, 28 mm Copper
13. Cable exits for power supply and sensors
14. Sub-cooler
15. Return line sensor
16. Condenser with drain for primary side
17. Compressor
18. Coolant in (to HP), 28 mm Copper
19. Coolant out (from HP), 28 mm Copper
20. Exit for communication cable

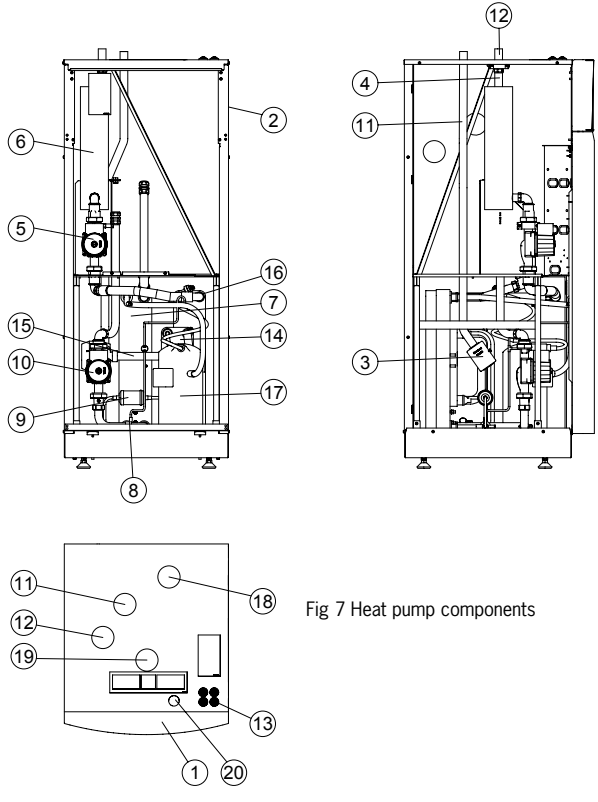


Fig 7 Heat pump components

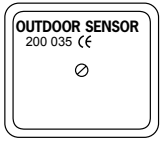
6 Outdoor temperature sensor

In order to measure the outdoor temperature as accurately as possible on buildings that are up to three storeys high, the sensor should be located 2/3 of the way up the wall of the building. On higher buildings, the sensor should be located between the second and third floors. The area where it is mounted must not be sheltered from the wind, nor exposed to direct wind. The outdoor sensor should not be mounted on a reflective sheet metal surface.

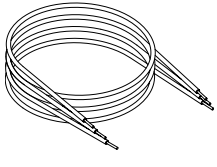
The sensor must be located at least 1 m from any openings in the wall of the building through which warm air can escape.If the sensor cable is connected through
a pipe, the pipe must be sealed so that the sensor is
not affected by any escaping air. Then connect the sensor to the heat pump control system as described
in the section on Electrical installation.



Room thermostat (option)



Outdoor sensor



Temperature sensor insert and tube

Conversion table for sensors

Before measuring the resistance of sensors the sensor cables must first be disconnected from the control equipment.

First measure sensor with cable.

Then measure sensor alone.

Outdoor

°C	Ohm
-30	1884
-25	1443
-20	1115
-15	868
-10	681
- 5	538
0	428
5	343
10	276
15	224
20	183
25	150
30	124
35	103
40	86

Supply/Return/Warm water
Supply/Return/Coolant out

°C	kOhm
0	66,3
5	52,4
10	41,8
15	33,5
20	27,1
25	22,0
30	18,0
35	14,8
40	12,2
45	10,1
50	8,5
55	7,1
60	6,0
65	5,0
70	4,2
75	3,7
80	3,1
85	2,7
90	2,3
95	2,0

7 Electrical installation

7.1 Electrical connections

Electrical installation may only be carried out by an authorized electrician (and must comply with the relevant local and national regulations). Electrical installation must take the form of a permanently wired connection. It must be possible to isolate the mains supply by means of an all-pole switch (circuit breaker) with a contact opening of at least 3 mm.(The maximum load for externally connected units is 2 A).

Electric current!
The connectors carry electrical current and can pose a risk to life due to electric shock. The mains supply must be disconnected before starting work on electrical installation. The heat pump is connected internally at the factory. Electrical installation work therefore involves the following main steps:

Connecting to the mains electricity supply

- Remove the front panel from the heat pump.
- Run the supply cable through the opening provided in the top panel of the heat pump to the terminal blocks.
- Connect the supply cable to the terminal block (fig 8).

Note that the supply cable must only be connected to the designated ter minal block. Do not connect to any other terminal blocks!

Note that the max. power of the additional heating must be selected, see page 45.

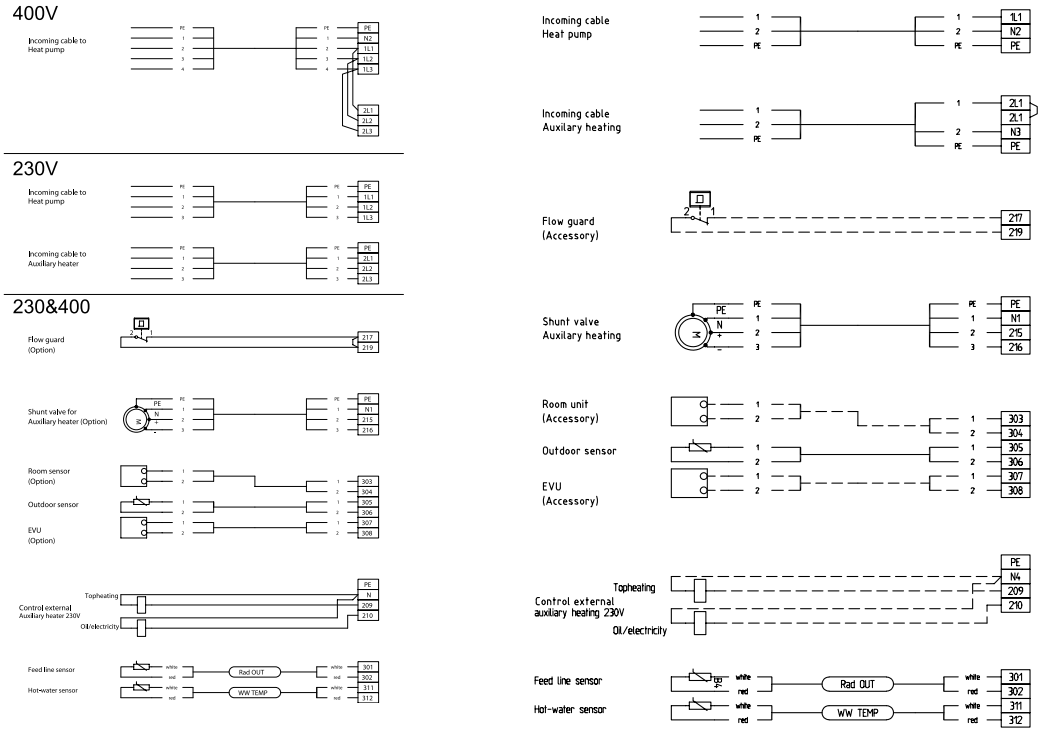
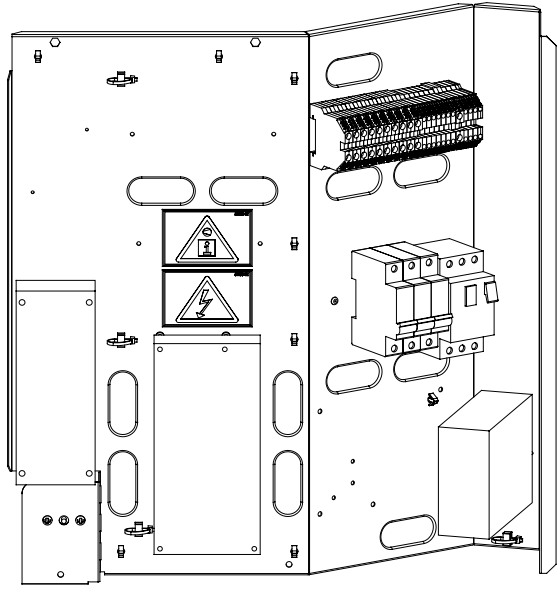


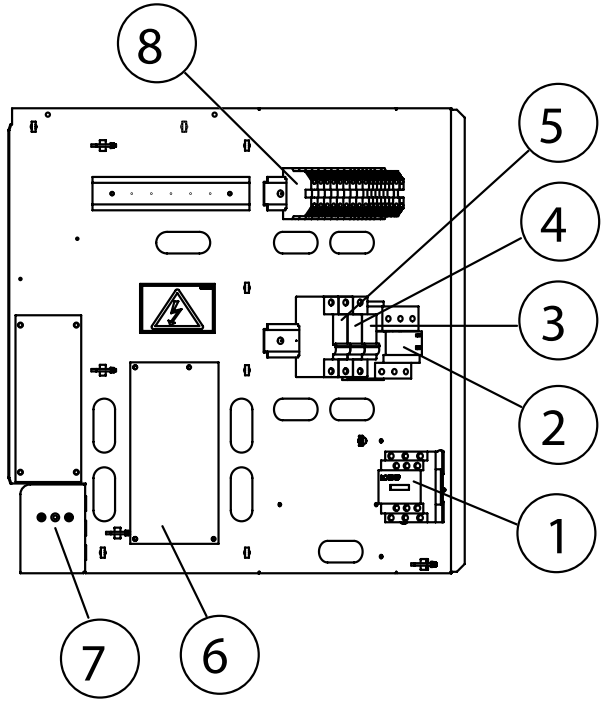
Fig. 8: Electrical connections

Electrical cubicle 55527-B



1. K1 Contactor compressor
2. F11 Overcurrent relay, compressor
3. F2 Operating fuse, circulation pump
4. F1 Operating fuse, brine pump
5. F0 Operating fuse, switching valve
6. Control computer
7. T1 Thermal cut-out, electric heater
8. Terminal blocks for incoming supply and temperature sensor

Electrical cubicle 53369-A



1. K1 Compressor starter
2. F11 Overcurrent relay, compressor
3. F2 Operating fuse, circulation pump
4. F1 Operating fuse, brine pump
5. F0 Operating fuse, switching valve
6. Control computer
7. T1 Thermal cut-out, electric heater
8. Terminal blocks for incoming supply and temperature sensor
9. Terminal blocks for warm water, supply line and operating additional heater

7.1.1 Connecting the outdoor temperature sensor

Remove the front panel from the heat pump.

Run the connecting cable for the outdoor sensor through the opening provided in the top panel to the terminal block.

Connect the sensor to the terminals shown in fig. 9 page 21.

- Note that the outdoor sensor uses Safety Extra-Low Voltage (SELV) connections.
- Please refer to the special installation instructions for the outdoor temperature sensor

7.1.2 Connecting the room thermostat (option)

Remove the front panel from the heat pump.

Run the room thermostat cable to the terminal block through the opening provided in the top panel.

Connect the sensor to the terminals shown in fig. 9 page 21.

- Note that the room thermostat uses Safety Extra-Low Voltage (SELV) connections.
- Please refer to the special installation instructions for the room thermostat!

7.1.3 Connecting the supply line sensor 230V

The supply line sensor is installed and connected after the internally mounted electric heater. If the heating system has an externally mounted additional heater (electric, oil-fired, etc.) the sensor must be placed after this heater's shunt (if fitted). Connect to terminal numbers: 301 & 302 according to wiring diagram.

Note that the supply line sensor uses Safety Extra-Low Voltage (SELV) connections.

Choice of heating system
When delivered the heat pump is preset for use with a VL heating system (with built-in electric heater). It can be reset for use with a D system by using the control computer (see Control panel).

System VL (see page 11) – operated in parallel with internal electric heater.

The internal electric heater is connected and its power output can be restricted in steps of 1,5, 3 or 4,5 kW (see Control panel).

System VL – operated in parallel with external electric heater without shunt.

If an external electric heater is used together with an internal power rating of more than 4,5kW the control for the additional external heater must be connected to terminal numbers: neutral and 210. System VL must be selected. The supply line sensor is then moved out and placed after the external additional heater. The control panel is programmed using the service menu – Add. heater Max step "lightning".

System D (see page 12) – operated in parallel with other external additional heaters (electric boiler or oil-fired boiler with shunt)

The shunt valve is connected to earth on the following terminal blocks. Neutral Shunt +215 and shunt –216. Control for additional external heaters is connected to terminal number: 210. The supply line sensor is moved out and placed after the shunt. The control panel is programmed using the service menu – Add. heater to Max step 1H (switch-off delay 1 h).

System D – operated in parallel with external electric heater.

The internal electric heater is disconnected by disconnecting the connectors from the supply cables. The connectors must then be insulated from accidental contact. Control for the external electric heater is connected to terminal number: 210. The control panel is programmed using the menu – Add. heater Max step "lightning". If an electric heater is required for top heating, this is connected to terminal number: 209. Note! This must be installed separately in the final warm water tank. The interval time is set via the control panel, using the service menu – Warm water. A separate thermostat must be used to the control the stop temperature at which top heating is switched off.

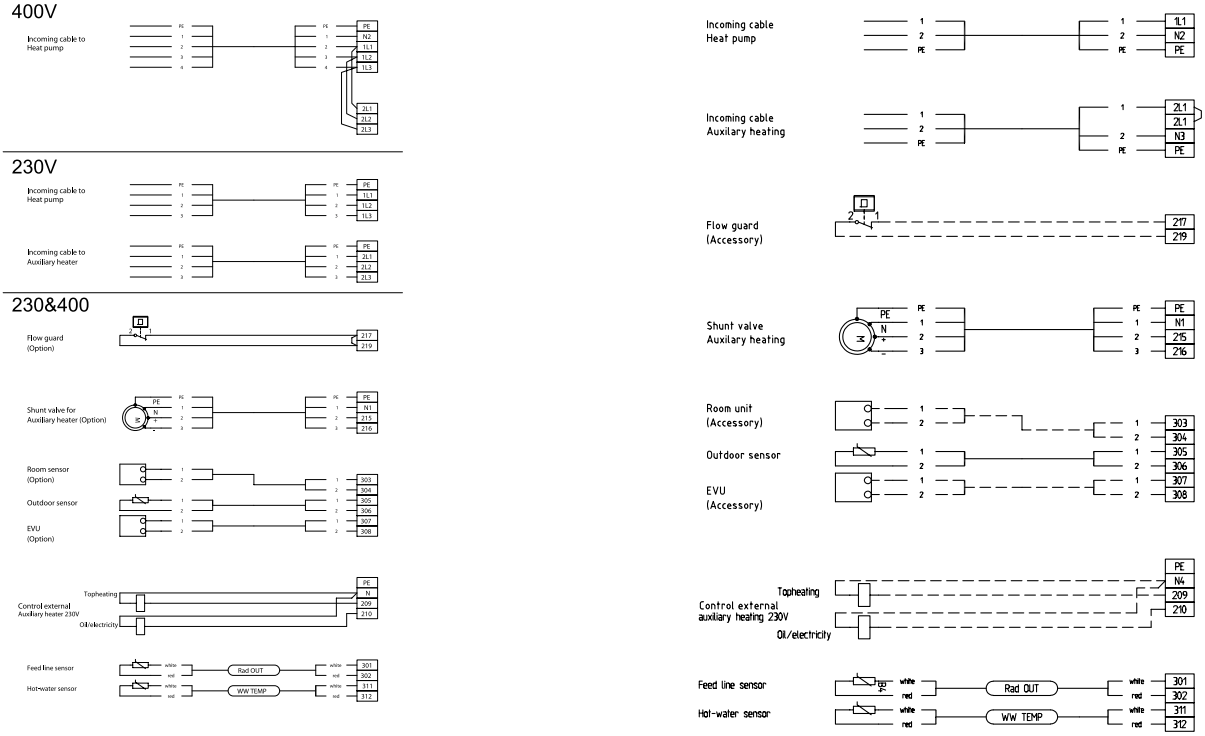


Fig. 9. Connection of control equipment and external sensors

Connecting the supply line sensor (cont)

The supply line sensor is installed and connected after the internally mounted electric heater. If the heating system has an externally mounted additional heater (electric, oil-fired, etc.) the sensor must be placed after this heater's shunt (if fitted). Connect to terminal numbers: 301 & 302 according to wiring diagram.

Note that the supply line sensor uses Safety Extra-Low Voltage (SELV) connections.

Choice of heating system
When delivered the heat pump is preset for use with a VL heating system (with built-in electric heater). It can be reset for use with a D system by using the control computer (see Control panel).

System VL – operated in parallel with internal electric heater, delivering up to 9 kW rated output.

The internal electric heater is connected and its power output can be restricted in steps of 3, 6 or 9 kW (see Control panel).

System VL (see page 11) – operated in parallel with external electric heater without shunt, delivering more than 9 kW rated output.

If an external electric heater is used together with an internal power rating of more than 9kW then control for the additional external heater must be connected to terminal numbers: neutral and 210. System VL must be selected. The supply line sensor is then moved out and placed after the external additional heater. The control panel is programmed using the service menu – Add. heater Max step "lightning".

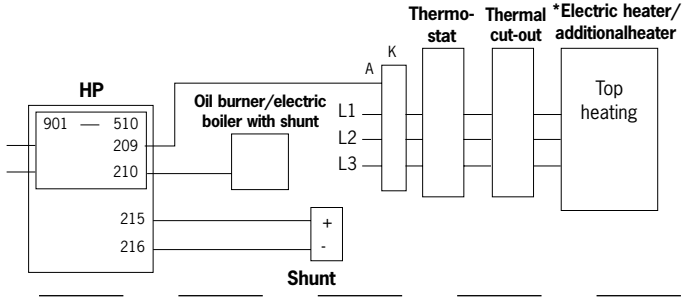
System D (see page 12) – operated in parallel with other external additional heaters (electric boiler or oil-fired boiler with shunt)

The shunt valve is connected to earth on the following terminal blocks. Neutral Shunt +215 and shunt –216. Control for additional external heaters is connected to terminal number: 210. The supply line sensor is moved out and placed after the shunt. The control panel is programmed using the service menu – Add. heater to Max step 1H (switch-off delay 1 h).

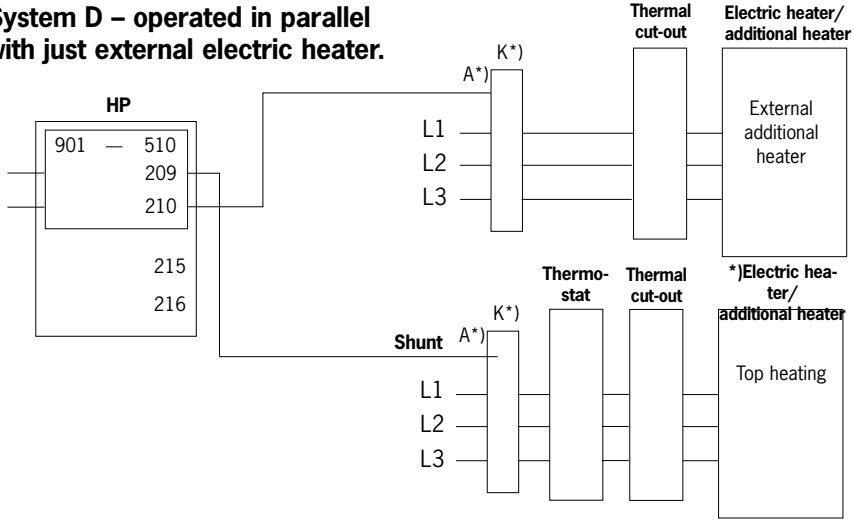
System D (see page 12) – operated in parallel with external electric heater.

The internal electric heater is disconnected by disconnecting the connectors from the supply cables. The connectors must then be insulated from accidental contact. Control for the external electric heater is connected to terminal number: 210. The control panel is programmed using the menu – Add. heater Max step "lightning". If an electric heater is required for top heating, this is connected to terminal number: 209. Note! This must be installed separately in the final warm water tank. The interval time is set via the control panel, using the service menu – Warm water. A separate thermostat must be used to the control the stop temperature at which top heating is switched off.

System D – operated in parallel with other external heaters (electric boiler or oil-fired boiler with shunt)



System D – operated in parallel with just external electric heater.



*) The electric heater is intended for top heating in a warm water tank. The interval time for top heating is set via the control panel, using the top heating menu. A separate thermostat must be used to control the stop temperature at which top heating is switched off, as shown in the figure.

K*) Contactor for electric heater.

A*) Connection for operating contactor.

7.2 Additional features

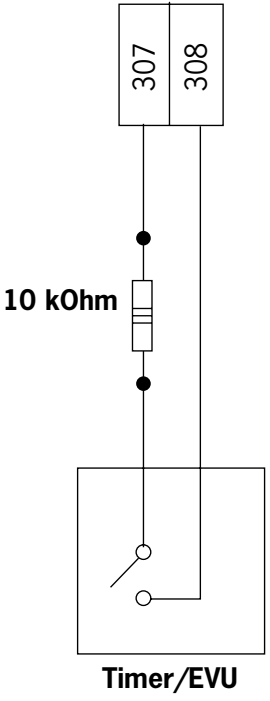
When terminal numbers 307 and 308 are connected together it enables the EVU (Elektrizitäts-Versorgungs-Unternehmen) function.

This feature prevents operation of the heat pump, additional heating and circulation pump as long as the terminals remain connected.

The text "EVU STOP" appears in the display.

Connecting terminals 307 and 308 with a 10 kohm resistor enables the function "ROOM REDUCTION".

The size of the reduction is set using the menu "Heat curve".



8 Filling the coolant system

The coolant pump must be running in order to fill the coolant system, which means that electrical installation must be complete.

The recommended antifreeze is ethylene glycol or ethanol with corrosion inhibitor, diluted to a concentration of 30 percent by volume. The collector hose, DN40, holds about 1.0 litre/metre. Note that it is important for the correct operation and reliability of the installation that the collector hose is bled fully when the installation is brought into use.

– Switch the heat pump "OFF" (see "Selecting operating mode").
– Use an external container (66) (plastic bowl etc.) to mix water and antifreeze in the correct proportions.

Note that each batch must be well mixed. The system is then filled with the coolant mixture from the bowl with the aid of an external pump fitted with a strainer with a maximum particle size of 1 mm (67) that is capable of bleeding the collector hose (66), as shown on the following page. Connect the pressure line from the pump to the filler connection near the shut-off valve (62).

– Close the shut-off valve (57).
– Open the shut-off valve (62) and valve (64).
– Open the shut-off valve (61) and connect back to the bowl with a hose.
– Start the external pump and fill the collector hose, then start the coolant pump (29) as well by selecting the manual service menu from the control panel. Note! Switched off after around 10 min.
– Let the coolant pump (29) run so that the air in the system collects in the bleed tank.

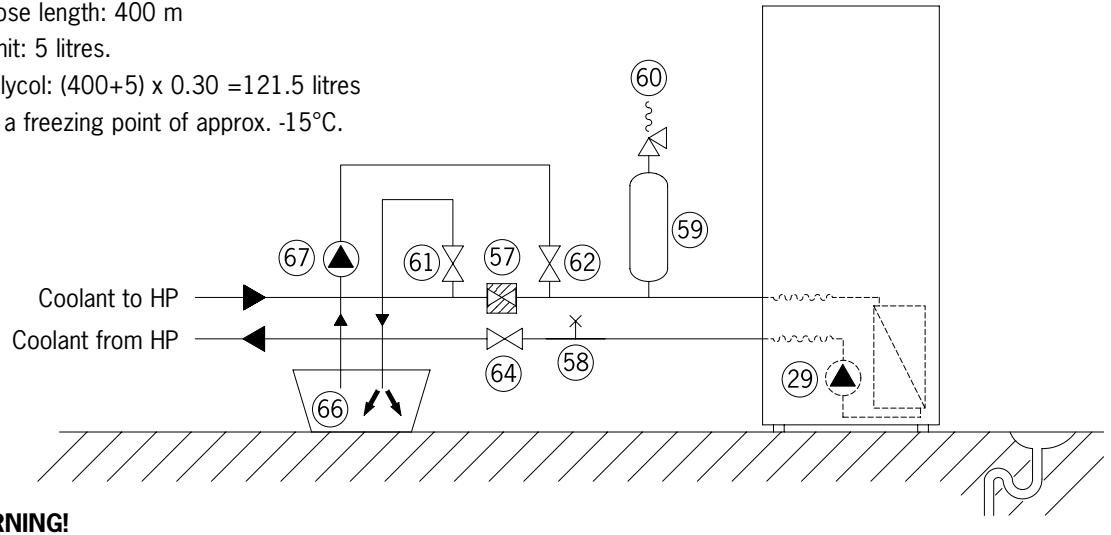
  As the air collects in the bleed tank the level of liquid in the tank will drop. Fill as described above.
– Open the valve (60) and release any excess pressure. The liquid level should not fall below 2/3 of the tank height.

  Check that the valves (61 and 62) are closed.
– Switch the control panel from off, manual to the chosen operating mode if the heating system has been filled.

Filling the coolant system

Coolant volume:
(hose length + 5 x 0.30) = volume of antifreeze.

Example: Hose length: 400 m
Volume in unit: 5 litres.
Volume of glycol: (400+5) x 0.30 =121.5 litres
which gives a freezing point of approx. -15°C.



WARNING!
Ethylene glycol and ethanol (alcohol) must be handled in accordance with the instructions on their containers!

9 Preparing for use

Do not start the equipment until the heating system, warm water tank (if fitted) and coolant circuit have been filled. Otherwise the pumps may be damaged.

If the installation is only to be operated in additional heating mode, first make sure that the heating system has been filled and that neither the coolant pump nor the compressor can start.

This is done by switching the operating mode to "Add. heater".

Follow the safety precautions!

The following points must also be checked before using the equipment for the first time:

- Check that all connections are tight.
- Check the direction of rotation of the compressor by starting the compressor. If the direction of rotation is incorrect this will be indicated by the text "Err. phase seq." on the control computer display.

  If this is the case it will be necessary to swap two of the phases on the incoming supply (e.g. L1 and L3).
- Adjust the factory settings to suit the customer's requirements.

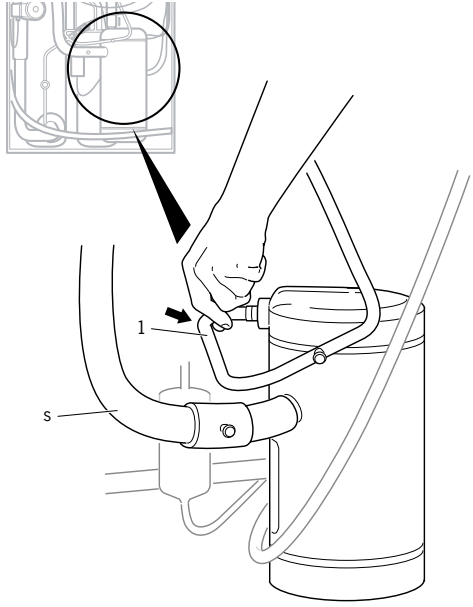


Fig. 10. Check the direction of rotation of the compressor and change it if necessary.

9.1 Fitting the front panel

Fit the front panel as follows:

- Open the hinged top panel (1).
- Locate the slot (2) at the bottom of the unit and push the panel gently upwards so that it covers the front.
- Tighten the screws (3).

Take care to avoid damaging the front panel!

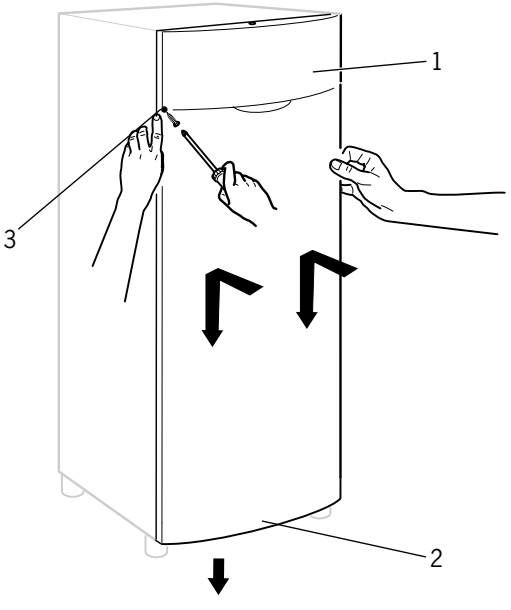


Fig. 11. Fitting the front panel

9.2 Requirements for switching on heating

9.2.1 Heat pump (compressor)

- The integral value is less than the value set in "Integral HP", and
- more than 5 minutes have elapsed since the heat pump stopped, and
- the interval since the heat pump was last started, as set by "HP interval", has elapsed, and
- the control pressostat is closed (=1), and
- if the outdoor temperature is less than the heat stop temperature.

9.2.2 Additional heating

- The integral value is less than the value set in "Integral HP + Integral TS (additional heater)", and
- the actual supply line temperature is more than 3°C below the set supply line temperature, and
- the control pressostat is closed (=1).

9.3 Requirements for switching off heating

9.3.1 Heat pump (compressor)

- The integral value is higher than 0°C min., or
- the return line temperature is greater than or equal to the set maximum return line temperature, or
- if the outdoor temperature is higher than the heat stop temperature.

9.3.2 Additional heating

- The integral value is higher than the set value for "Integral HP", or
- the actual supply line temperature is more than 3°C above the set supply line temperature.

9.4 Requirements for switching on warm water

9.4.1 Heat pump (compressor)

- The warm water temperature is equal to or lower than the set value for warm water start, or
- the warm water temperature is up to 2°C higher than the set value for warm water start when heating is switched off (so-called Top-up).

9.4.2 Additional heating

Additional heating is not normally used to produce warm water, except in the following situations with a VL system:

1. The additional heater was already switched on when warm water operation began.
2. During top heating (legionella function). The additional heater is switched on 2 minutes after the start of top heating.
3. When operating mode "additional heater only" is chosen.

9.5 Requirements for switching off warm water

9.5.1 Heat pump (compressor)

- When the control pressostat has been open for more than 5 seconds (40 seconds within the first two minutes of starting warm water operation), or
- if heating is required at the same time, the system alternates between heating and warm water operation according to the times set by "Time heat" and "Time warm water", or
- if the maximum time of 45 minutes has elapsed; after this time the heating demand is checked for 5 minutes.

9.5.2 Additional heating

For VL systems, heating is switched off when:

- the set value for Top H. stop is reached, or
- when the "Add. heater" operating mode is selected and the temperature is equal to the "Warm water stop" value.

10 Computer control panel

The Thermia Diplomat control panel consists of a character display, four control keys and an alarm indicator. Next to the control panel are care instructions, brief instructions on how to increase and decrease the indoor temperature, and a decal giving the name and telephone number of Thermia's dealer.

10.1 Features

The control computer is operated by means of a user-friendly menu system that appears in the character display. There are four different sub-menus that you reach from a master menu. You can find more information about the different menus a little further on in these instructions.

To select the menus and to increase or decrease the set values, you use the four control keys:

- An up key with plus sign
- A down key with minus sign
- A right key
- A left key

The alarm indicator at the bottom of the control panel shows whether the installation is operating (steady glow) or whether an alarm has been activated (flashing light). The indicator can be seen clearly even when the hinged panel has been lowered over the control panel.

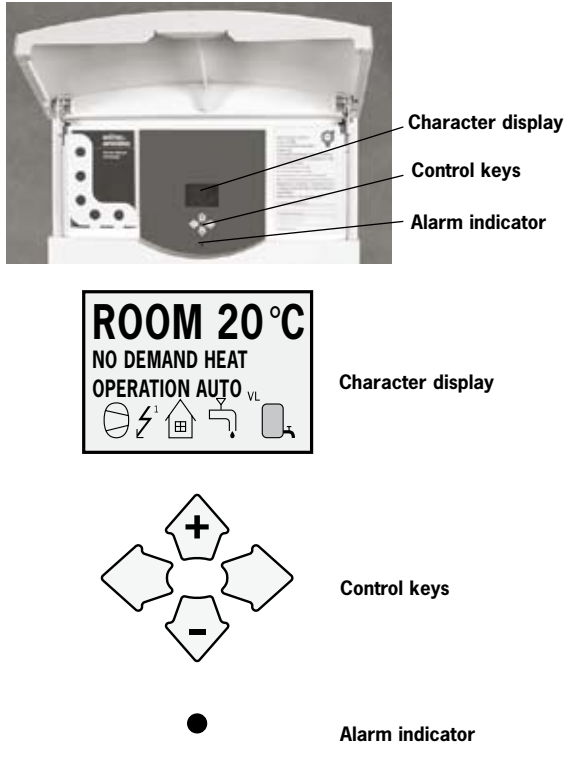


Fig. 14. Computer control panel

10.2 Display information and settings

10.2.1 Normal

Shows text and operating status. If a room thermostat is not fitted the heating can be adjusted using the +/– arrows. Returns after 10 seconds. If there is an alarm this is displayed 10 seconds after the last key press.

Message	Meaning
ROOM	Shows desired temperature. If a room thermostat is connected it shows the set temperature and actual temperature.
ERR. PHASE SEQ.	Incorrect phase sequence to compressor. Show only on display and only for first 10 mintes.
HIGH RETURN	Return temperature too high for heat pump.
HEATPUMP	Displayed if demand for HP.
START	Displayed if demand for HP and no start delay.
START_MIN	Displayed if start delay set for HP.
ADD.HEATER	Displayed if demand for additional heater.
EVU STOP	Input closed and everything switched off.
COOLING	Displayed if passive cooling is being used.
COOLING A	Displayed if active cooling is being used.

Display information and settings – Normal (cont.)

Message	Meaning
HP SYMBOL 	Displayed if compressor is in operation. F is displayed if flow monitor is installed. 0 is displayed if operating pressostat is activated.
LIGHTNING SYMBOL 	Displayed if ADD.HEATER ON. Figure shows number of steps.
HOUSE SYMBOL 	Displayed if house is heated by compressor or electric heater.
COOLING SYMBOL 	Displayed if cooling is on. A indicates active cooling.
TAP SYMBOL 	Displayed if warm water operation is switched on.
SYSTEM	Shows selected system VL / D / VL + F / D + F.
CLOCK SYMBOL 	Displayed if reduction is switched on.
TANK SYMBOL 	Displayed if warm water is switched on. The tank is filled at the warm water temperature. Empty tank shows set start temperature. Full tank shows temperature when last filled (51°C at start), but max. 54°C and min. 49°C. Set stop temperature is shown if additional heating is used. During charging the tank begins filling at the current temperature. Lightning symbol next to tank indicates top heating.

10.2.2 Alarms

Displayed in the event of an alarm. Alarms 1–5 can be reset by switching OPERATION = OFF or switching the power supply off and on again. ***Only applies to Thermia Duo Optimum**

Message	Meaning
HIGHPRESS ERROR	High pressure error. Compressor is switched off. No warm water production.
LOWPRESS ERROR	Low pressure error. Compressor is switched off. No warm water production.
MOTOR P ERROR	Motor cut-out triggered (overcurrent relay for compressor). Compressor is switched off. No warm water production.
BRINE OUT	Brine temperature is not above set value. Compressor is switched off. No warm water production.
	Not normally activated.
BRINEFLOW	Flow monitor not active during last start attempt. Compressor is switched off. No warm water production.
ADD.HEATER	Thermal cut-out triggered. No additional heater active.
SENSOR OUTDOOR	Outdoor sensor fault. 0°C degrees is used.
SENSOR SUPPLY	Supply line sensor fault. Everything is switched off except radiator pump.
SENSOR RETURN	Return sensor fault. Return temp. = Supply line temp -5°C is used. Max. estimated supply line temp. 45°C.
SENSOR WARMWATER	Warm water sensor fault. No warm water production.
CIRC. PUMP*	The circulation pump's integrated alarm has deployed
BRINE PUMP*	The brine pump's integrated alarm has deployed. Both these alarms are reset by holding the red button on the respective pump depressed for at least 1 second.

10.2.3 CUSTOMER INFORMATION

Customer settings. Changes can be made to operating mode and curves. Current temperatures and operating times, plus history which provides information about installation. From the "Normal" menu, press the right or left key to go to "Information".

10.2.3.1 Operation

Selecting heat source

Message	Meaning
OFF	Installation switched off. Alarms are reset.
AUTO	Heat pump and additional heater can be used. AUTO is displayed if additional heater is switched off.
HEATPUMP	Heat pump can be used.
ADD.HEATER	Additional heater can be used.
WARMWATER	Warm water production only, using heat pump and additional heater for top heating.
MANUAL TEST	Outputs are controlled manually. Cannot be adjusted here.

10.2.3.2 Heat curve

Heat curve settings.

Message	Meaning
CURVE	Temperature when outdoor temperature is 0°C.
MIN	Minimum supply line temperature if no heat stop.
MAX	Maximum supply line temperature.
CURVE5	Change in slope of curve when outdoor temperature reaches +5°C.
CURVEA0	Change in slope of curve when outdoor temperature reaches 0°C.
CURVE-5	Change in slope of curve when outdoor temperature reaches -5°C.
HEATSTOP	Maximum outdoor temperature at which heating is allowed.
REDUCTION	Reduction in set room temperature. Activated when 10 kohm resistor is connected across EVU input.
ROOM FACTOR	Factor that is multiplied by room temperature difference, and affects supply line temperature. Displayed if room thermostat is connected.

10.2.3.3 Heat curve 2

Displayed if sub-shunt sensor is connected (option).

CURVE2	Sub-shunt temperature when outdoor temperature is 0°C.
MIN	Minimum sub-shunt temperature if no heat stop.
MAX	Maximum sub-shunt temperature.

10.2.3.4 Temperature

Displays current temperatures, history and set/calculated values. History is shown over the selected time interval (see right side). In the event of an alarm the history is not logged for 1 week or until the alarm is reset by switching Operation = Off.

OUTDOOR	0°C	Right arrow shows history.
ROOM	20°C	
SUPPLYLINE	41(40)°C	Supply line temperature (desired temperature for house). Right arrow shows history.
RETURNLINE	34(48)°C	Return line temperature (set stop temperature). Right arrow shows history.
WARMWATER	44(44)°C	Warm water temperature. Right arrow shows history.
INTEGRAL	-40	Calculated integral. Right arrow shows history.
BRINE OUT	-3°C	Outgoing brine temperature. Right arrow shows history.
BRINE IN	0°C	Incoming brine temperature. Right arrow shows history.
COOLING	15(16)°C	Expansion card. Temperature for shunt cooling (set temperature). Option.
SUBSHUNT	33(35)°C	Expansion card. Temperature sub-shunt (desired temperature). Option.
CURRENT	17(20)A	Expansion card. Installation current (set by main fuse). Option.

10.2.3.5 Operating time

HEATPUMP	Operating time for compressor
ADD.HEATER	Operating time for additional heater rated at 3 kW (VL).
ADD.HEATER 2	Operating time for additional heater rated at 6 kW (VL).
WARMWATER	Operating time for warm water production by compressor.
COOLING	Total time (active and passive). Displayed if cooling sensor is connected.
COOLING A	Operating tome for active cooling. Displayed if cooling sensor is connected and NODE >19.

10.2.4 SERVICE

Menus for installation and testing. Obtained by holding in right and left arrow keys for 5 seconds.

10.2.4.1 Warm water

START	Start temperature for warm water production. OFF = no sensor alarm.
TIME WARMWATER	Time allocated to warm water when demand is shared (minutes).
TIME HEAT	Time allocated to heating when demand is shared (minutes).
TOPH. INTERVAL	Interval in days between top heating. 1 = constant..
TOPH. STOP	Stop temperature for top heating (VL).

10.2.4.2 Heat pump

Settings for heat pump.

INTEGRAL A1	Start value integral for heat pump.
HYSTERESIS	Forced start for heat pump. Integral is A1.
MAX RETURN	Stop temperature when return temperature too high.
STARTINTERVAL	Minimum interval between restarting heat pump (minutes).
ALARM BRINE	Alarm temperature for outgoing brine, which stops heat pump.
SHUNT COOLING	Temperature for shunt cooling. Temperature to prevent condensation (Expansion card).
LIQUID LINE	Sensor for testing.

10.2.4.3 Additional heater

Settings for additional heater.

INTEGRAL A2	Start value for integral +A1 for additional heater (and 2°C under-temperature).
HYSTERESIS	Forced start for additional heater. Integral is A1+A2.
MAX STEP	Maximum step for additional heater. 0 = no additional heater / 1=3 kW / 2=6 kW / 3=9 kW
MAX CURRENT	Main fuse for installation. Applies only if circuit board with power relay is fitted, for example.
SHUNTTIME	Interval time for shunt motor. Applies to all connected sub-shunts. Time specifies minimum interval between signals.
WW STOP	Stop temperature for Operation ADD.HEATER (VL).

10.2.4.4 Manual test

Used to test outputs.

Message	Meaning
MANUAL TEST 0	0 = display outputs, 1 = switch to manual, 2 = manual to check temperatures.
HEATPUMP	Start heat pump. Also starts brine pump. Pump cannot be started if an alarm is active.
BRINEPUMP	Output for brine pump.
CIRC.PUMP	Output for radiator pump.
VXV WARMWATER	Output for warm water switching valve.
ADD.HEATER 3	Output for immersion heater 3 kW (VL).
ADD.HEATER 6	Output for immersion heater 6 kW (VL).
ADD.HEATER	Output for oil (D).
SHUNT 1	Output for shunt (D).
TOPH. WARMWATER	Output for top heating (D).
ALARM	Output for external alarm (Expansion card).
SHUNT COOLING	Output for shunt cooling (Expansion card).
SUBSHUNT	Output for sub-shunt (Expansion card).
COOLING P	Output for relay, passive cooling (Expansion card).
COOLING A	Output for relay, active cooling (Expansion card).
0–10 V	Output 0–10V (Expansion card).

10.2.4.5 Installation 1A

Settings during installation. Version number for control card = 1. Version number for display = A.

Message	Meaning
ENGLISH	Chosen language
SERVICETIME	1 = speeds up integral calculation and start delays by a factor of 60.
FACTORY SET	Restores factory settings. R = Radiator, G = Floor.
RESET OPERATE TIME	Resets all operating times to zero.
CALIBRATION SENSOR	The following sensors are included in the heat pump: OUTDOOR, SUPPLYLINE, RETURNLINE, WARMWATER, BRINE OUT and BRINE IN.
SYSTEM	Selection of system type: VL or D, and with flow monitor (F).
LOGGTIME	Time in minutes between recordings of temperature history (if no alarm is active).
NODE 1	Displayed if interface is connected.
BRINETIME ON	Switch-on delay for brine pump.
BRINETIME OFF	Switch-off delay for brine pump.
TOPH.TIME	Activates 3 hours of top heating. ON = active (VL). Setting: see installation 1 A.
TOPH.TIME D	Time for top heating (D).

10.2.4.6

This menu is only available in Thermia Duo Optimum

Message	Meaning
CIRC: DIFF	Setting the desired temperature difference in the heat transfer system.
BRINE. DIFF	Setting the desired temperature difference in the coolant system.
FLOW RAD	AUTO = Automatic adjustment of the temperature difference in the heat transfer system. 3 - 10 = manually set speed.
FLOW BRINE	AUTO = Automatic adjustment of the temperature difference in the coolant system. 3 - 10 = manually set speed.
MAX HW	Setting the highest desired supply line temperature during hot water heating.
MIN HW	Setting the lowest desired supply line temperature during hot water heating.

10.3 Control system – available operating modes

The control computer's "OPERATION" menu lets you choose between the following five operating modes:

- OPERATION OFF: The system is switched off completely. The text "OPERATION ⏻" appears in the control panel's character display.

  Remember that if the installation is switched to "OPERATION OFF" for long periods in the winter the heating system must be drained of water, otherwise there is a risk of pipes bursting due to freezing.
- OPERATION AUTO: The heat pump and additional heater are controlled automatically by the control computer. The text "OPERATION AUTO" appears in the control panel's character display.
- OPERATION HEATPUMP: The control computer only allows the heat pump unit (compressor) to operate. The text "OPERATION HEATPUMP" appears in the control panel's character display.
- OPERATION ADD.HEATER: The control computer only allows the additional heater to operate. This operating mode can be used when a new installation is brought into use, if the coolant system is not ready for operation. The text "OPERATION ADD. HEATER" appears in the control panel's character display.
- OPERATION WARMWATER: In this mode the heat pump only produces warm water; it does not provide any heat to the heating system. The text "OPERATION WARMWATER" appears in the control panel's character display.

To change the operating mode:

1. Press the right or left key once to select the main "INFORMATION" menu. The cursor will highlight the menu selection "OPERATION".
2. Open your choice by pressing the right key once. The cursor will highlight the operating mode that was used last.
3. Select the new operating mode using the up or down keys.
4. Press the left key twice to return to the character display's operating mode.

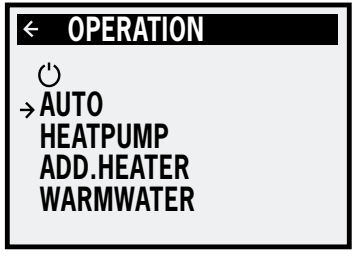


Fig. 15. Operating modes

10.4 Heating system

10.4.1 Adjusting the "ROOM" setting

You can also adjust the heat curve and hence the indoor temperature by changing the "ROOM" setting. If the "ROOM" setting is used to adjust the heat curve it does not make the curve steeper or shallower, as happens when the "CURVE" value is changed. It instead shifts the entire heat curve through 3°C for each one degree change in the "ROOM" setting.

The relationship between the supply line temperature and the outdoor temperature is unaffected, since the supply line temperature is increased or decreased by the same amount over the full length of the curve. In other words the entire heat curve is raised or lowered, instead of changing its slope.

This method of adjusting the indoor temperature should only be used to raise or lower the indoor temperature temporarily. If a long-term increase or decrease in the indoor temperature is required you should adjust the heat curve instead, as this is the most energy-efficient and cost-effective way of setting the heat curve (see "Adjusting the "CURVE" setting" on the following page). The factory-set "ROOM" setting is 20°C.

10.4.2 To change the "ROOM" setting:

1. Press the up or down key once to select the "ROOM" setting so that you can change it.
2. Increase or decrease the "ROOM" setting using the up or down keys to give the desired indoor temperature. Wait 10 seconds or press the left key once to return to the character display's operating mode.

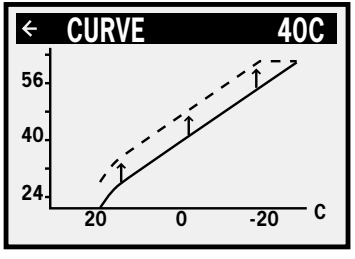


Fig. 16. Setting the ROOM value

10.5 Setting the heat curve

10.5.1 Adjusting the "CURVE" setting

You can adjust the heat curve by changing the "CURVE" setting. The "CURVE" setting determines the supply line temperature for the radiators when the outdoor temperature is 0°C. When the outdoor temperature is below 0°C the temperature of the supply line water that is piped to the radiators is raised above 40°C, and when the outdoor temperature is above 0°C the temperature of the supply line water is reduced below 40°C. When you increase the "CURVE" setting it makes the slope of the heat curve steeper, and when the setting is reduced it makes the slope shallower.

This method of adjusting the indoor temperature should be used to make a long-term change to the temperature, as this is the most energy-efficient and cost-effective method. A temporary increase or decrease can be made by adjusting the "ROOM" setting instead (see "Adjusting the "ROOM" setting on the following page").

The factory setting is 40°C at an outdoor temperature of 0°C. The setting can be adjusted between 22 and 56°C.

10.5.2 To change the "CURVE" setting:

1. Press the right or left key once to select the main "INFORMATION" menu. The cursor will highlight the menu selection "OPERATION".
2. Press the down key to move the cursor over the menu selection "HEATCURVE".
3. Open the menu by pressing once on the right key. The cursor will highlight "CURVE".
4. Open your selection by pressing the right key once.
5. Increase or decrease the setting using the up or down keys.

The graph will show how the slope of "CURVE" changes.

Press the left key three times to return to the character display's operating mode.

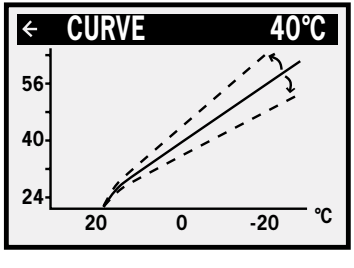


Fig 17. Changing the CURVE setting

10.6 Maximum return line temperature for heating system

The maximum return line temperature is set at the factory at 48°C. This setting must be adapted to suit each individual installation. In order to enter a new setting the heat pump must be supplying the heating system and in steady operation.

The maximum return line temperature setting is changed from the "Service Menu" under "HEATPUMP".

- Work out the difference between the set values for the supply line temperature and return line temperature, in this case 36 – 30 = 6.
- Subtract this value from 55°C (maximum permitted supply line temperature), i.e. 55 – 6 = 49.
- Enter the new value of 49°C in the "Service Menu" under "HEATPUMP".

10.7 Adjusting the room factor

The room factor is only displayed if a room thermostat is installed. The room factor affects the heat curve and can be set between 0 (= no effect) and 4 (= maximum effect).

For underfloor heating we recommend a setting in the range 1–3, and for radiator heating a setting in the range 2–4.

This setting is entered using the "Heat curve" menu and is called the "ROOM FACTOR".

10.8 Warm water production

The water temperature in the warm water tank is controlled by the control pressostat and cannot be adjusted.

10.8.1 Reading the warm water temperature

The current warm water temperature can be read from the character display by following these steps:

1. Press the right or left key once to open the main "INFORMATION" menu.

   The cursor will highlight the menu selection "OPERATION".

2. Press the down key to move the cursor to the menu selection "TEMPERATURE".
3. Open this menu by pressing the right key once.
4. Press the down key to move the cursor down to "WARMWATER".
5. Open this menu by pressing the right key once.

   A graph will be displayed showing the warm water temperature over the last few hours.

6. Press the left key three times to return to the character display's operating mode.

The value that is shown by the "WARMWATER" menu is the current warm water temperature, and the value in brackets is the temperature at which warm water production starts.

When the temperature of the warm water drops below the start value the installation will start producing warm water.

11 Troubleshooting

Alarms

If an error message appears, try restarting the equipment by switching the circuit breaker on and off.
If restarting the heat pump does not work, try to solve the problem with the aid of the table below.

Message	Meaning	Cause	Remedy
LOWPRESS ERROR	Pressure is too low – The compressor is switched off and no warm water is produced.	Not enough antifreeze in the coolant system. Air on the coolant system. Blocked particle filter in coolant system.	Fill with coolant containing the correct percentage of antifreeze, and bleed through the safety valve on the coolant tank. Check and clean the particle filter if necessary.
		Coolant pump has stopped or is running too slow.	Measure the temperature difference between the incoming and outgoing coolant. The difference should be between around 3–5°C.
		Ice formation in coolant system.	Measure the percentage of antifreeze, wait until the ice melts, which prevents the low pressure pressostat from closing the electrical circuit.
		Cooling system inadequately filled or leaking.	Drain and search for leak, then refill with new refrigerant. Do not refill with used coolant.
		Clogged or saturated drying filter in cooling circuit.	Measure temperature difference – max. value 2°C.

Troubleshooting – alarms (cont.)

Message	Meaning	Cause	Remedy
HIGHPRESS ERROR	Pressure is too high – The compressor is switched off and no warm water is produced.	Radiator or underfloor heating thermostats are not turned up high enough. Air in heating system. Clogged particle filter in heating system, located in heat pump return line.	Turn up radiator or underfloor heating thermotat. Top up and bleed heating system. Measure temperature difference between heat pump supply and return lines.
		Refrigerant circuit over-filled.	Drain refrigerant circuit and refill with fresh refrigerant according to instructions.
		Heating pump has stopped or is running too slow.	Check heating pump.
MOTOR P ERROR	Motor cut-out has tripped (compressor overcurrent relay). The compressor is switched off and no warm water is produced.	Loss of phase or tripped fuse.	Check fuses and cable connections (power supply to compressor).
		Loose electrical connections.	Check electrical connections.
		Mains voltage too low.	Measure voltage supplied heat pump electrical connections.
ERR PHASE SEQ.	Compressor rotating in wrong direction, no warm water is produced.	Incorrect phase sequence to compressor.	Change phase sequence by switching two phases over (at power supply terminals).

Troubleshooting – alarms (cont.)

Message	Event
SENSOR ERROR	Outdoor sensor: 0°C is displayed (flashing) in the menu for outdoor temperature and is used to calculate the set temperature for the supply line. **Supply line sensor:** The control computer is switched to the OFF setting and the circulation pump for the heating system is running. **Return line sensor:** The return temperature is calculated as follows: supply line temp. -5°C (-15°C two minutes after warm water production begins). **Warm water sensor:** No warm water production. **Room sensor:** Actual room temperature is not displayed. 20°C is used to calculate set supply line temperature.
ALARM-	Alarms are reset by switching off the circuit breaker, or by switching the control computer to the **RESET** OFF/Standby setting. When there is an alarm the graphic display of temperatures is locked to simplify troubleshooting. When the control computer is switched to "OFF/Standby" the temperatures are shown on the graphic display again.

12 Factory settings for control computer

Parameter	Factory setting	Min.	Max.
ROOM	20°C	10°C	30°C
OPERATION AUTO	(OFF when delivered)		
HEATCURVE			
CURVE	40°C (Floor=30)	22°C	56°C
MIN	10°C	10°C	50°C
MAX	55°C (Floor=45)	40°C	70°C
CURVE 5	0°C	-5°C	5°C
CURVE 0	0°C	-5°C	5°C
CURVE -5	0°C	-5°C	5°C
HEATSTOP	17°C	0°C	40°C, OFF
REDUCTION	3°C	1°C	10°C
ROOM FACTOR	2	0	4
HEATCURVE 2 Only when expansion card installed			
CURVE2	40°C	22°C	56°C
MIN2	10°C	10°C	40°C
MAX2	55°C	30°C	70°C

Factory settings for heat pump control computer (cont.)

Parameter	Factory setting	Min.	Max.
WARMWATER			
START	44°C	OFF, 30card	46card
TIME WARMWATER	20 MIN	5 MIN	40 MIN
TIME HEAT	20 MIN	5 MIN	40 MIN
TOPH. INTERVAL	14 D	OFF, 1 D	90 D
TOPH. STOP	65°C	50°C	65°C
HEATPUMP			
INTEGRAL (A1)	60	5	99
HYSTERESIS	7°C	1°C	15°C
MAX RETURN	48°C	30°C	55°C
STARTINTERVAL	20 MIN	1 MIN	30 MIN
ALARM BRINE	OFF	OFF, -14	10
SHUNT COOLING	16°C	0°C	30°C
ADD.HEATER			
INTEGRAL (A2)	600	50	990
HYSTERESIS	15°C	5°C	30°C
MAX STEP	2	OFF, 1	3
MAX CURRENT	20 A	16 A	35 A
SHUNTTIME	60 S	30 S	99 S
WARMWATER STOP	65°C	50°C	65°C

Factory settings for heat pump control computer (cont.)

Parameter	Factory setting	Min.	Max.
CALIBRATION SENSOR			
OUTDOOR	0°C	-5°C	5°C
SUPPLYLINE	0°C	-5°C	5°C
RETURNLINE	0°C	-5°C	5°C
WARMWATER	0°C	-5°C	5°C
BRINE OUT	0°C	-5°C	5°C
BRINE IN	0°C	-5°C	5°C
SYSTEM VL			
LOGGTIME	1 MIN	1 MIN	60 MIN
NODE	1	0	30
BRINETIME ON	30 S	10 S	600 S
BRINETIME OFF	30 S	10 S	600 S
TOPH. TIME	OFF	OFF	ON
TOPH. TIME D	1 H	1 H	10 H

13 Refrigerant

The compressor (refrigerant) circuit of the heat pump is charged with the refrigerant R404A for Duo 230V and R407C for Duo 400V.

Service and repair work on the refrigerant circuit must be carried out by certified refrigeration personnel only.

Flammability

R404A and R407C is neither flammable nor explosive.

Toxicity

Under normal circumstances, and used correctly, the refrigerant has a low level of toxicity. In spite of this, it may cause injury (even danger to life) under abnormal conditions or if deliberately misused.

Refrigerant gases are many times heavier than air. If there is a leak of refrigerant in a small and badly ventilated room, strong concentrations of refrigerant gas might form with resulting lack of oxygen and risk of suffocation. Therefore, the heat pump should always be placed in a room with good ventilation.

In connection with open flame the refrigerant forms phosgene, a pungent gas that is easily recognizable by its smell already in concentrations well below the allowed limit. In this case, evacuate the premises and air well.

Any person showing symptoms of intoxication must be moved into the fresh air immediately.

Service work on the refrigerant circuit

When repairs are made to the refrigerant circuit, no refrigerant may escape to the open air. All refrigerant must be destroyed according to regulations. Draining and filling of new refrigerant is made through the service valves. For required amount of new refrigerant please refer to the unit nameplate. If other refrigerant is used than R404A for Duo 230V and R407C for Duo 400V, the Thermia Värme warranty will be cancelled unless the new refrigerant was previously confirmed in writing by Thermia as being an approved replacement for R404A or R407C.

Scrapping

Before scrapping the heat pump, the refrigerant must be destroyed according to local and national health and environment regulations.

14 Technical specifications

NOTE! Although R407C is a chlorine-free refrigerant with low environmental impact and does not affect the ozone layer, work may only be carried out on the system by a suitably qualified person.

Heat pump, model:			***Duo 4***	***Duo 6***	***Duo 8***	***Duo 10***	***Duo 12***	***Duo 16***
Refrigerant	– type		R407C	R407C	R 407C	R407C	R407C	R407C
	– quantity	kg	0.75	1.1	1.25	1.6	1.7	1.8
	– test pressure	MPa	3.2	3.2	3.2	3.2	3.2	3.2
	– design pressure	MPa	2.9	2.9	2.9	2.9	2.9	2.9
Compressor	– type		Scroll	Scroll	Scroll	Scroll	Scroll	Scroll
	– oil type		Ester	Ester	Ester	Ester	Ester	Ester
Electrical data	Mains supply	voltage	400V 3-N	400V 3-N	400V 3-N	400V 3-N	400V 3-N	400V 3-N
	Power rating Compressor	kW	*	2	2,3	3,6	4,4	5,6
	Power rating Compressor + AH	kW	*	5.0[2]/8.0[3]/11.0[4]	5.3[2]/8.3[3]/11.3[4]	6.6[2]/9.6[3]/12.6[4]	7.4[2]/10.4[3]/13.4[4]	8.6[2]/11.6[3]/14.6[4]
	Immersion heater	kW	3/6/9	3/6/9	3/6/9	3/6/9	3/6/9	3/6/9
	Start current LRA	A	36	14	25	28	28	47
	Fuse	A	*	10[2]/16[3]/20[4]	16[2]/16[3]/20[4]	16[2]/16[3]/20[4]	16[2]/20[3]/25[4]	20[2]/20[3]/25[4]
Nominal power	Available power [1]	kW	3.76/3.67	5.78/5.52	8.13/7.73	9.54/9.24	11.12/10.53	16.61/15.78
	Heat factor [1]	COP	4.07/2.83	4.2/3.1	4.6/3.4	4.4/3.3	4.5/3.3	4.4/3.2
Nominal flow rate	– coolant	l/s	0.20	0.31	0.47	0.57	0.63	0.92
	– heating medium	l/s	0.09	0.13	0.19	0.22	0.26	0.4
External available pressure [5]	– coolant [5]	kPa	35	31	33	32	64	56
	– heating medium [5]	kPa	50	43	39	34	48	40
Duo Optimum	– coolant [5]	kPa	*	89	92	82	82	70
	– heating medium	kPa	*	109	106	99	97	88
Max/Min temperatures:	– coolant in	°C	20/-10	20/-10	20/-10	20/-10	20/-10	20/-10
	– heating medium	°C	55/20	55/20	55/20	55/20	55/20	55/20
Weight (excl. packaging)		kg	140	145	150	155	165	175

[1] According to EN 255 when coolant in/heating medium out temperatures are 0/35°C and 0/50°C (incl. circulation pumps).

[2] Heat pump with 3 kW additional heater

[3] Heat pump with 6 kW additional heater

[4] Heat pump with 9 kW additional heater

[5] The pressure drop that must not be exceeded outside the heat pump without the flow rate dropping below the nominal value. For coolants these values require coolant hose measuring 40 x 2.4.

*Data not available at date of printing. Contact Thermia Värme AB

Technical specifications (contd.)

NOTE! Although R407C is a chlorine-free refrigerant with low environmental impact and does not affect the ozone layer, work may only be carried out on the system by a suitably qualified person.

Heat pump, model:			*Duo 4 ES*	*Duo 6 ES*	*Duo 8 ES*	*Duo 10 ES*	*Duo 12 ES*	*Duo 16 ES*
Refrigerant	– type		R407C	R407C	R 407C	R407C	R407C	R407C
	– quantity	kg	0.75	1.1	1.25	1.6	1.7	1.8
	– test pressure	MPa	3.2	3.2	3.2	3.2	3.2	3.2
	– design pressure	MPa	2.9	2.9	2.9	2.9	2.9	2.9
Compressor	– type		Scroll	Scroll	Scroll	Scroll	Scroll	Scroll
	– oil type		Ester	Ester	Ester	Ester	Ester	Ester
Electrical data	Mains supply	voltage	230V 1-N	230V 1-N	230V 1-N	230V 1-N	230V 1-N	230V 1-N
	Power rating Compressor	kW	2,7	3,3	4,2	5,4	5,7	5,6
	Power rating Compressor + AH	kW	4,2[2]/5,7[3]/7,2[4]	4,8[2]/6,3[3]/7,8[4]	5.7[2]/7.2[3]/8.7[4]	6.9[2]/8.4[3]/9,9[4]	7.2[2]/8.7[3]/10,2[4]	8.6[2])/11.6[3])/14.6[4])
	Immersion heater	kW	1,5/3/4,5	1,5/3/4,5	1,5/3/4,5	1,5/3/4,5	1,5/3/4,5	3/6/9
	Start current LRA	A	36	58	76	97	108	47
	Fuse	A	45[2]/45[3]/45[4]	45[2]/45[3]/45[4]	45[2]/45[3]/45[4]	45[2]/45[3]/63[4]	45[2]/45[3]/63[4]	
Nominal power	Available power [1]	kW	3.76/3.67	5.78/5.52	8.13/7.73	9.54/9.24	11.12/10.53	16.61/15.78
	Heat factor [1]	COP	4.07/2.83	4.2/3.1	4.6/3.4	4.4/3.3	4.5/3.3	4.4/3.2
Nominal flow rate	– coolant	l/s	0.20	0.31	0.47	0.57	0.63	0.92
	– heating medium	l/s	0.09	0.13	0.19	0.22	0.26	0.4
External available pressure [5]	– coolant [5]	kPa	35	31	33	32	64	56
	– heating medium [5]	kPa	50	43	39	34	48	40
Duo Optimum	– coolant	kPa	*	89	92	82	82	70
	– heating medium	kPa	*	109	106	99	97	88
Max/Min temperatures:	– coolant in	°C	20/-10	20/-10	20/-10	20/-10	20/-10	20/-10
	– heating medium	°C	55/20	55/20	55/20	55/20	55/20	55/20
Weight (excl. packaging)		kg	140	145	150	155	165	175

[1] According to EN 255 when coolant in/heating medium out temperatures are 0/35°C and 0/50°C (incl. circulation pumps).

[2] Heat pump with1,5 kW additional heater

[3] Heat pump with3 kW additional heater

[4] Heat pump with 4,5 kW additional heater

[5] The pressure drop that must not be exceeded outside the heat pump without the flow rate dropping below the nominal value. For coolants these values require coolant hose measuring 40 x 2.4.

*Data not available at date of printing. Contact Thermia Värme AB

Technical specifications (contd.)

ATTENTION: R404A is an environmentally sound, non-chloric refrigerant that does not affect the ozone layer. Nevertheless all service work involving the refrigerant circuit must be carried out by a qualified technician.

Heat pump, model:			***Duo D 550***	***Duo D 750***	***Duo D 1050***	***Duo D 1550***	***Duo D1800***
Refrigerant:	- Type		R404A	R404A	R404A	R404A	R404A
	- Quantity	kg	1,4	1,7	1,8	2,25	2,5
	- Test pressure	MPa	3,2	3,2	3,2	3,2	3,2
	- Design pressure	MPa	2,9	2,9	2,9	2,9	2,9
Compressor	- Type		Scroll	Scroll	Scroll	Scroll	Scroll
	- Type of oil		Ester	Ester	Ester	Ester	Ester
Electrical data	Mains voltage	Volt	230V	230V	230V	230V	230V
	Auxiliary heater	kW	3/6/9	3/6/9	3/6/9	3/6/9	3/6/9
	Starting amps LRA	A	43–48	61–70	87–95	126–139	161-172
	Fuse auxiliary heater	A	10[2]/16[3]/25[4]	10[2]/16[3]/25[4]	10[2]/16[3]/25[4]	10[2]/16[3]/25[4]	10[2]/16[3]/25[4]
	Fuse heat pump		A	10	16	16	32 32
Nominal capacity:	- Capacity output [1]	kW	5.4/5.0	7.3/7.1	10.2/9.4	15.6/14.7	17.7/16.5
	- Heat factor [1]	COP	4.2/2.8	4.4/3.0	4.6/3.0	4.3/3.0	4.5/3.3
Nominal flow:	- Heating circuit	l/s	0,1	0,2	0,3	0,4	0,5
	- Chilled water circuit	l/s	0,3	0,5	0,6	0,9	0,95
Available pressure drop [3]	- Chilled water circuit[3]	kPa	50	41	30	50	40
	- Heating circuit[3]	kPa	37	19	42	125	102
Duo Optimum	- Chilled water circuit	kPa	*	*	*	*	*
	- Heating circuit	kPa	*	*	*	*	*
Max/Min temperatures:	- Chilled water circuit in	°C	20 / -10	20 / -10	20 / -10	20 / -10	20 / -10
	- Heating circuit	°C	55 / 20	55 / 20	55 / 20	55 / 20	55 / 20
Weight excl. packing		kg	140	145	150	165	175

[1] Provided (according to EN 255) entering chilled water / leaving chilled water 0 / 35°C and 0 / 50°C (including circulation pumps).

[2] Heat pump with 3 kW auxiliary heater.

[3] Heat pump with 6 kW auxiliary heater.

[4] Heat pump with 9 kW auxiliary heater.

[5] Pressure drop that must not be exceeded outside the heat pump without the nominal flow being reduced.

For the brine circuit these values require a water hose of Ø 40 x 2.4.

*Data not available at date of printing. Contact Thermia Värme AB

Installation and commissioning have been carried out by:

As a user you should ensure that the installer fills in the following information in order to facilitate any service requirements.

PIPE INSTALLATION

Date . . . . . Company . . . . . .

Name . . . . . Tel. no. . . . . .

ELECTRICAL INSTALLATION

Date . . . . . Company . . . . . .

Name . . . . . Tel. no. . . . . .

BALANCING OF SYSTEM

Date . . . . . Company . . . . . .

Name . . . . . Tel. no. . . . . .

If these instructions are not followed during installation, operation and maintenance, the obligations of Thermia Värme AB under the warranty regulations are no longer binding.



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